Scheda iMX6: derivata da nanoberry di vpecanins (github.com/EnzoRF/nanoberry), GPLv3. Modifiche: via display, touch, espansione, hub USB e CP2102N; DDR3L 1 GB; USB-C alimentazione/OTG + USB-C host; console su header; bus MIKILAB; Wi-Fi/BT. Alimentazioni e segnali tra fogli sono etichette globali.

DDR3L

File: ddr.kicad_sch

SoC alimentazioni

File: soc-alimentazioni.kicad_sch

SoC periferiche 1

File: soc-periferiche-1.kicad_sch

SoC periferiche 2

File: soc-periferiche-2.kicad_sch

PMIC PF1550

File: pmic.kicad_sch

Ethernet

File: ethernet.kicad_sch

Boost batteria

File: boost.kicad_sch

Clock e controllo

File: clock.kicad_sch

Alimentazione bus e USB-C

File: alimentazione-bus.kicad_sch

USB host e console

File: usb-host.kicad_sch

Bus MIKILAB

File: bus-mikilab.kicad_sch

Radio

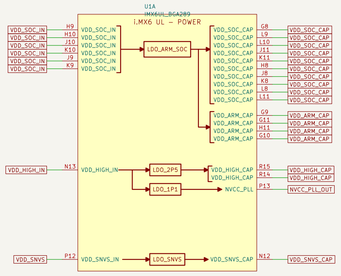
File: radio.kicad_sch

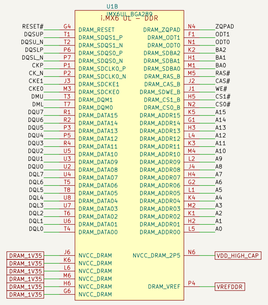
Sheet: /
File: iMX6.kicad_sch

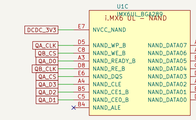
Title:

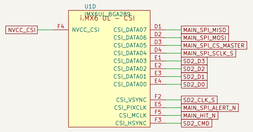
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/13

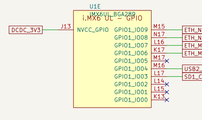
U2 DDR3L 8 Gb x16 = 1 GB: Alliance AS4C512M16D3LC-12BIN, FBGA-96 9x13 mm, stessa griglia JEDEC della W634GU6MB di nanoberry. A15 (M7) era gia' collegata e sbroglliata.

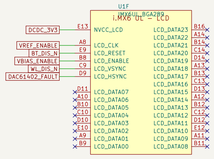


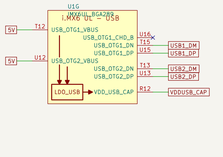


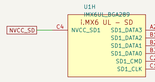


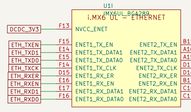


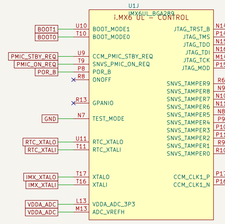


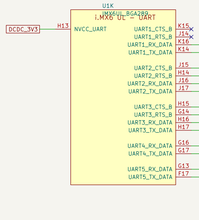


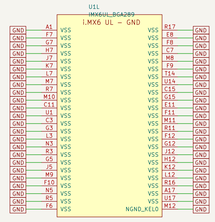


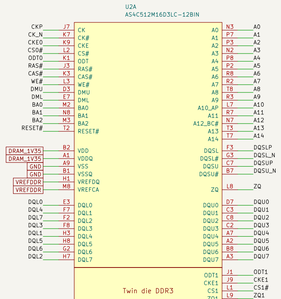


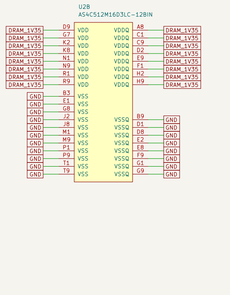


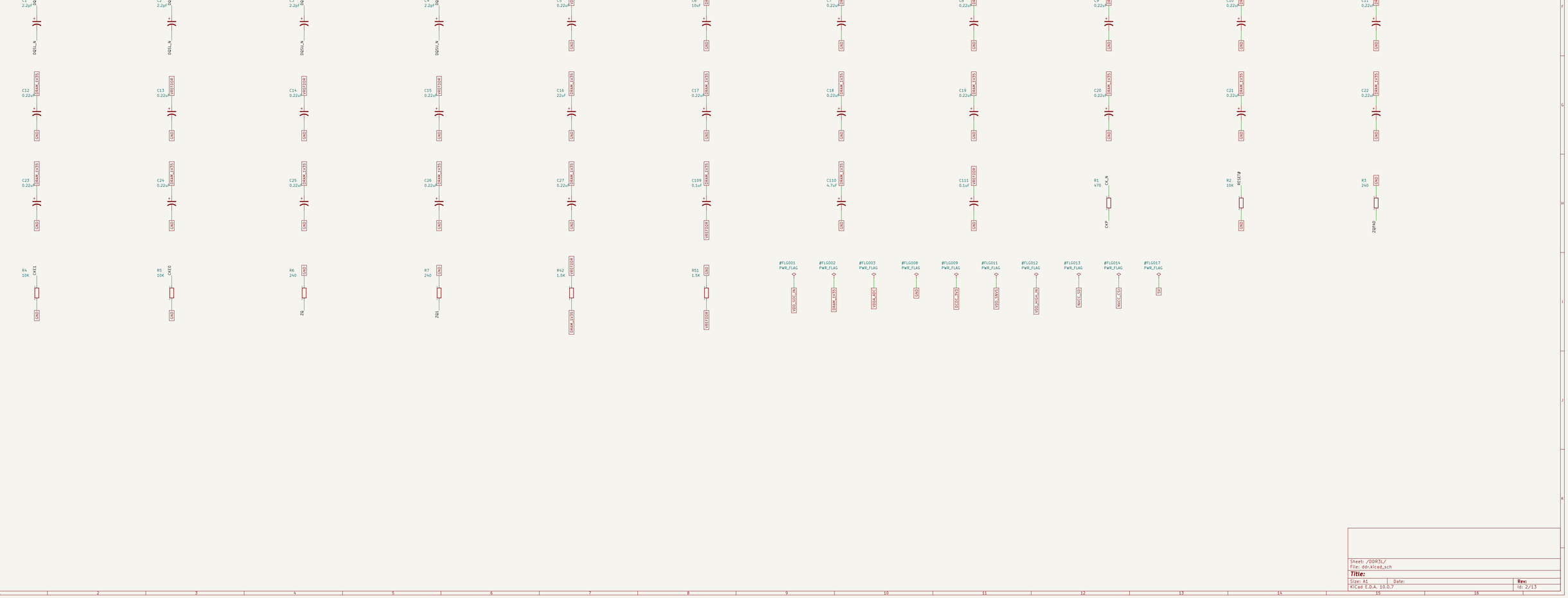








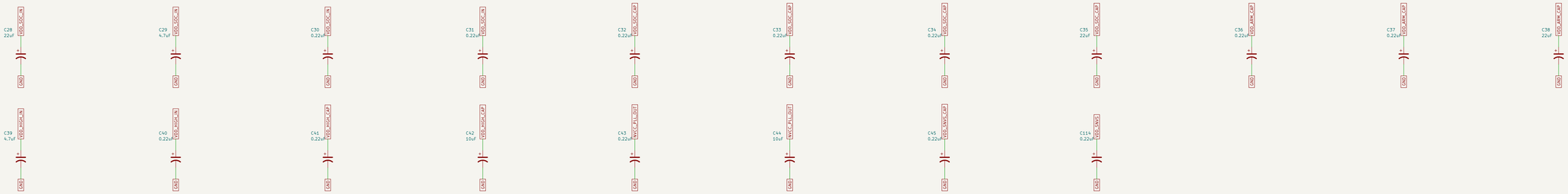




Sheet: /DDR3L/
File: ddr.kicad_sch

Title:

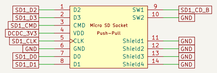
Size: A1 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 2/13

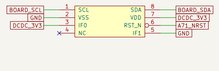


Sheet: /SoC alimentazioni/
File: soc-alimentazioni.kicad_sch

Title:

Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/13













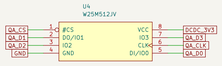


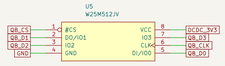




Sheet: /SoC periferiche 1/
File: soc-periferiche-1.kicad_sch
Title:
Size: A1 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 4/13















Sheet: /SoC periferiche 2/
File: soc-periferiche-2.kicad_sch
Title:
Size: A1 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 5/13

BT1
CR2032
LICELL
GND

D3
LED
CHGB
N_D3_Pad2

J2
PMIC_I2C
BOARD_SDA
BOARD_SCL
DCDC_3V3
VDD_SNVS
POR_B

J7
LiPo_THM
VBATT
BATT_THM
GND

Q1
2SA3018
STANDBY
N_Q1_Pad1
PMIC_STBY_REQ

U6
PF1550
DCDC
LDO
CHARGER
Only PF1550 (PF1510 alt too)
SW1IN
SW2IN
SW3IN
VLDO1IN
VLDO2IN
VLDO3IN
VINREFDDR
LICELL
VBUSIN(VIN)
USBPHY
INT2P7(LDO2P7)
THM(GND)
WDI
SDA
SCL
VDDIO
VDDOTP
VCORE
VDIG
SW1LX
SW1FB
SW2LX
SW2FB
SW3LX
SW3FB
VLDO1
VLDO2
VLDO3
VREFDDR
VSNVS
(GND)CHGB
VSYS1
VSYS2
(NC)VBATT1
(NC)VBATT2
POWERON
STANDBY
ONKEY
INTB
RESETBMCU
EP
VSYS
DRAM_1V35
N_R13_Pad2
BATT_THM
PMIC_WDI
BOARD_SDA
BOARD_SCL
DCDC_3V3
VDDOTP
VCORE
VDIG
5V
CHGB
VBATT
VSYS
SW1LX
SW1OUT
SW2LX
SW2OUT
SW3LX
SW3OUT
NVCC_CSI
VDD_HIGH_IN
NVCC_SD
VREFDDR
VDD_SNVS
CHGB
VSYS
VSYS
VBATT
VBATT
POWERON
STANDBY
ON_KEY
PMIC_INT
POR_B
GND

C48 4.7uF VSYS GND
C49 4.7uF VSYS GND
C50 4.7uF VSYS GND
C51 1.0uF VSYS GND

C52 4.7uF NVCC_CSI GND
C53 10uF SW2OUT GND
C54 10uF SW2OUT GND
C55 10uF SW2OUT GND
C56 10uF SW2OUT GND
C57 10uF SW1OUT GND
C58 10uF SW1OUT GND
C59 1.0uF VSYS GND
C60 4.7uF VDD_HIGH_IN GND
C61 1.0uF VSYS GND
C62 4.7uF NVCC_SD GND

C63 1.0uF DRAM_1V35 GND
C64 1.0uF VREFDDR GND
C65 0.1uF LICELL GND
C66 0.47uF VDD_SNVS GND
C67 100uF VSYS GND
C68 100uF VSYS GND
C69 4.7uF VBATT GND
C70 4.7uF VSYS GND
C71 1uF VCORE GND
C72 1uF VDIG GND

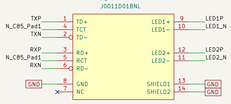
L1 1uH SW1OUT SW1LX
L2 1uH SW2OUT SW2LX
L3 1uH SW3OUT SW3LX

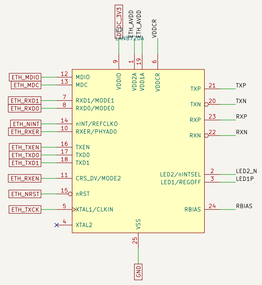
R8 0.02 SW1OUT VDD_SOC_IN
R9 0.02 SW2OUT DRAM_1V35
R10 0.02 SW2OUT DCDC_3V3
R11 47 N_D3_Pad2 VSYS
R12 DNP CHGB GND
R13 4.7K BATT_THM N_R13_Pad2
R14 DNP GND BATT_THM
R15 100K DCDC_3V3 PMIC_WDI
R16 4.7K DCDC_3V3 BOARD_SDA

R17 4.7K DCDC_3V3 BOARD_SCL
R18 100K ON_KEY VBATT
R19 100K PMIC_INT VDD_SNVS
R20 0R GND VDDOTP
R21 100K POR_B VDD_SNVS
R22 100K VDD_SNVS STANDBY
R23 100K VDD_SNVS N_Q1_Pad1
R24 DNP STANDBY PMIC_STBY_REQ
R25 100K GND N_Q1_Pad1
R41 0R POWERON PMIC_ON_REQ

#FLG004 PWR_FLAG VDDOTP
#FLG005 PWR_FLAG VCORE
#FLG006 PWR_FLAG VDIG
#FLG007 PWR_FLAG VBATT
#FLG010 PWR_FLAG VSYS

Sheet: /PMIC PF1550/
File: pmic.kicad_sch
Title:
Size: A1 Date: Rev:
KiCad E.D.A. 10.0.7 Id: 6/13

















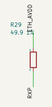










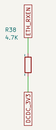














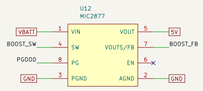




Sheet: /Ethernet/
File: ethernet.kicad_sch

Title:

Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 7/13















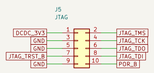


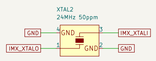


Sheet: /Boost batteria/
File: boost.kicad_sch

Title:

Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 8/13





















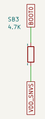














Sheet: /Clock e controllo/
File: clock.kicad_sch

Title:

Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 9/13

VIN_SYS 12 V dal BUS_POWER -> TPS56637 5 V -> TPS2121 IN1 (priorita'). IN2 = VBUS della USB-C 1 (dopo il polyfuse F2 di nanoberry). Uscita -> 5V di nanoberry (VBUSIN del PF1550).

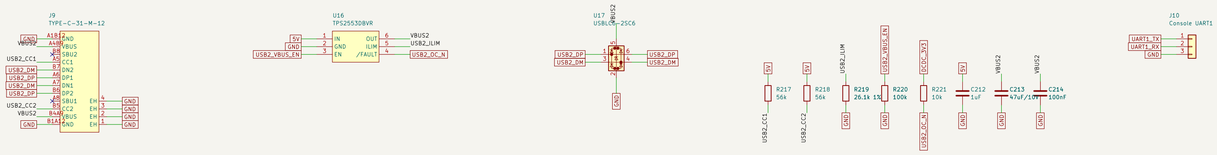
Il boost MIC2877 di nanoberry resta sul 5V (alimentazione da LiPo); il TPS2121 blocca la corrente inversa. Circuiti come sulle schede MIKILAB (SchedaIMX6ULL-SD).

U13
TPS56637RPAR
BUCK_EN
BUCK_FB
BUCK_PG
GND
MODE
PGND
VIN
BOOT
SW
VIN_SYS
BUCK_BST
BUCK_SW
L11
3.3uH/8A
5V_BUCK
TPS56637: EN 169k/36.1k, FB 73.2k/10k = 4.99 V, FF 20k + 100 pF
C201
10uF/25V
C202
10uF/25V
C203
100nF/50V
R201
169k 1%
R202
36.1k 1%
R203
100k
C204
100nF
R204
73.2k 1%
R205
10k 1%
R206
20k
C205
100pF
C206
47uF/10V
C207
47uF/10V
U14
TPS2121RUXR
5V_BUCK
MUX_PR1
MUX_OV1
VBUS1
MUX_OV2
MUX_SS
MUX_ILM
PWR_SRC_ST
TPS2121: PR1 4.49 V, OV 5.98 V, ILM 22.1k, SS 100 nF: PWR_SRC_ST con 10k
R207
32.4k 1%
R208
10k 1%
R209
46.4k 1%
R210
10k 1%
R211
46.4k 1%
R212
10k 1%
R213
22.1k 1%
C208
100nF
C209
1uF/10V
C210
1uF/10V
C211
47uF/10V
R214
10k
5V
CC/CC_PU2
J8
TYPE-C-31-M-12
DTG_VBUS
USB1_CC1
USB1_DN
USB1_DP
USB1_CC2
GND
VBUS
SBU2
CC1
DN2
DP1
DN1
DP2
SBU1
CC2
SHELL
U15
USBLC6-2SC6
VIN_SYS
USB1_DP
USB1_DM
USB-C 1: sink 5.1k su CC1/CC2
R215
5.1k
R216
5.1k
USB1_CC1
USB1_CC2

USB-C 2 host su USB_OTG2 (ex hub), affiancata alla USB-C 1. VBUS dal TPS2553 (~1 A, RILIM 26.1k). EN su GPIO1_IO04 con 100k a GND. FAULT su SNVS_TAMPER6 con 10k. Rp 56k su CC (sorgente default).

Console Linux su UART1 (al posto del CP2102N di nanoberry): header 1x3 TX, RX, GND a 3.3 V.

USB-C 2 host

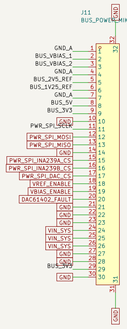


Sheet: /USB host e console/		
File: usb-host.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 11/13

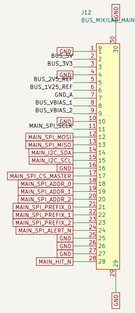
J BUS_POWER 30 pin (Hirose FH12-30S) su ECSPI4 (pad ENET2, ex touch). J MAIN 28 pin REV5.1 (KUNPU FPC-05F-28PH20) su ECSPI1 (pad CSI_DATA04-07, ex espansione P1). Mappa dei connettori come SchedaIMX6ULL-SD.

Serie 33R sui clock SPI. Pull-up 4.7k su ALERT#/HIT#, 2.2k su I2C. VREF_ENABLE 10k e VBIAS_ENABLE 100k a GND. CS e FAULT con i pull-up interni del 6ULL. ESD TPD4E05U06DQA sulle linee logiche.

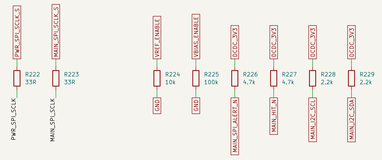
J11 — BUS_POWER_MIKILAB



J12 — BUS_MIKILAB_MAIN

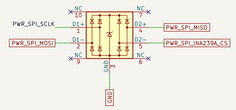


Serie SPI 33R, solo clock

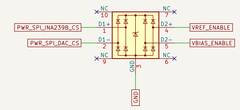


Livelli di riposo

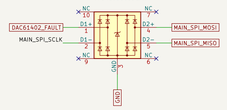
U18 — TPD4E05U06DQAR



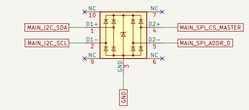
U19 — TPD4E05U06DQAR



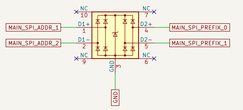
U20 — TPD4E05U06DQAR



U21 — TPD4E05U06DQAR



U22 — TPD4E05U06DQAR



U23 — TPD4E05U06DQAR

Disaccoppiamento

Flag

Sheet: /Bus MIKILAB/
File: bus-mikilab.kicad_sch

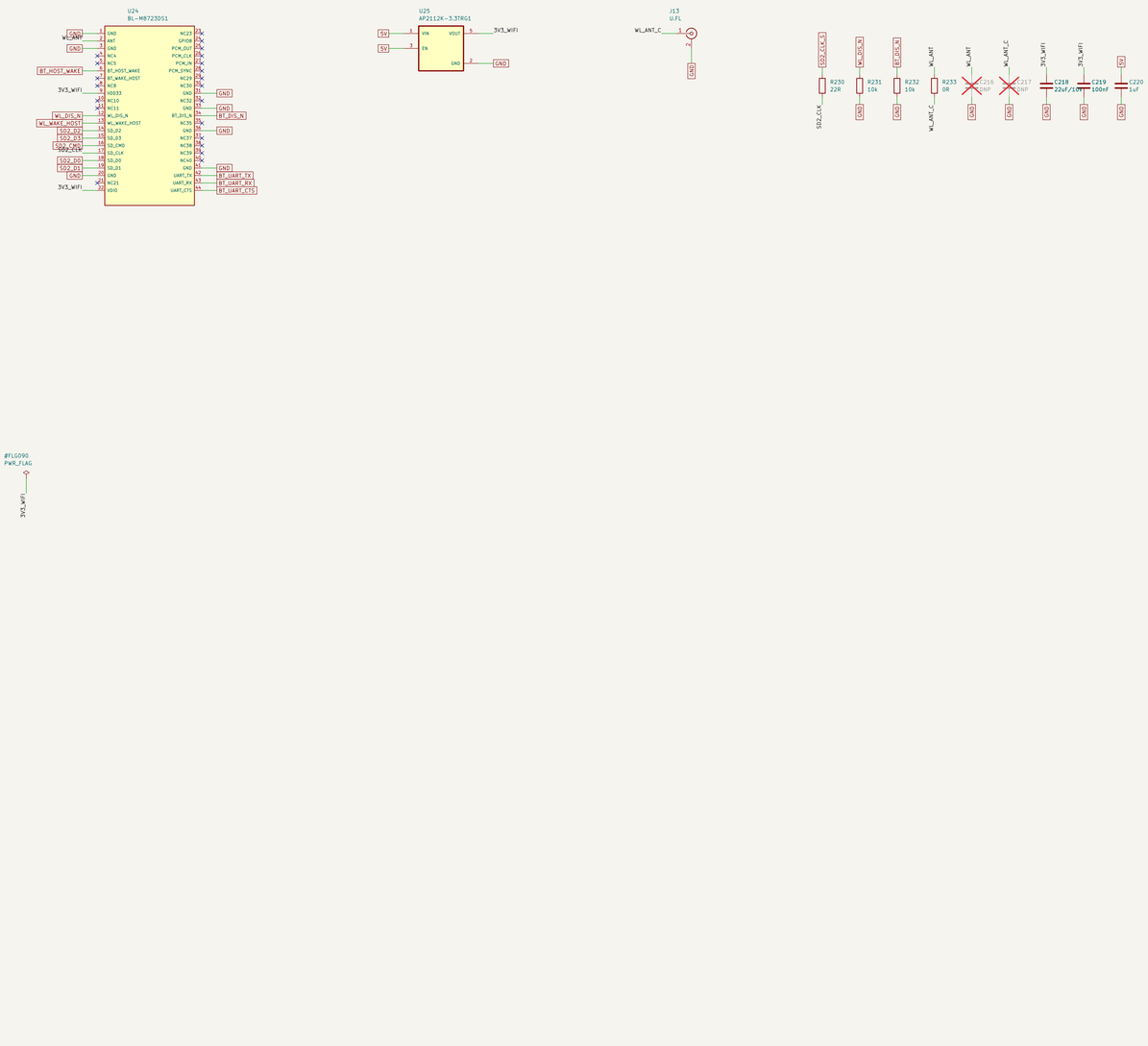
Title:

Size: A1 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 12/13

BL-M8723DS1 (RTL8723DS) su uSDHC2 sul pad CSI (VSYNC = CLK, HSYNC = CMD, DATA00–03). Libero perche' il boot e' da QSPI; BT su UART2 (TX, RX, RTS -> CTS del modulo). SDIO con i pull-up interni del 6ULL.

Alimentazione 3.3 V dedicata (AP2112K da 5V). WL_DIS# e BT_DIS# con 10k a GND. Antenna su U.FL con rete pi' OR + 2 DNP. Footprint LGA-44 dal datasheet LB-LINK V1.0.1.3.

Wi-Fi: 22R su CLK, enable a GND, rete pi



Sheet: /Radio/
File: radio.kicad_sch

Title:

Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 13/13